File Edit Selection View Go Run Terminal Help

Search

counttotalnumberoflinesque.py

C: > Users > Mo.Sajid > counttotalnumberoflinesque.py

```python
# File paths
input_filename = "input.txt"
output_filename = "output.txt"

# 1. Read all lines from input.txt
with open(input_filename, "r") as infile:
    lines = infile.readlines()

# 2. Count the total number of lines
total_lines = len(lines)

# 3. Extract the first two lines
first_two_lines = lines[:2]

# 4. Write the extracted lines to output.txt
with open(output_filename, "w") as outfile:
    outfile.writelines(first_two_lines)

# 5. Print the required messages to the screen
print(f"Total number of lines: {total_lines}")
print(f"First two lines written to {output_filename}")
```

Problems Output Debug Console Terminal Ports

Code

```
PS C:\Users\Mo.Sajid>
```

0 0

Ln 21, Col 55 Spaces: 4 UTF-8 CRLF { } Python Python 3.14.3

ENG IN 12:31 26-10-2026

File Edit Selection View Go Run Terminal Help

Search

counttotalnumberoflinesque.py

C: > Users > Mo.Sajid > counttotalnumberoflinesque.py

```python
# File paths
input_filename = "input.txt"
output_filename = "output.txt"

# 1. Read all lines from input.txt
with open(input_filename, "r") as infile:
    lines = infile.readlines()

# 2. Count the total number of lines
total_lines = len(lines)

# 3. Extract the first two lines
first_two_lines = lines[:2]

# 4. Write the extracted lines to output.txt
with open(output_filename, "w") as outfile:
    outfile.writelines(first_two_lines)
```

Problems Output Debug Console Terminal Ports

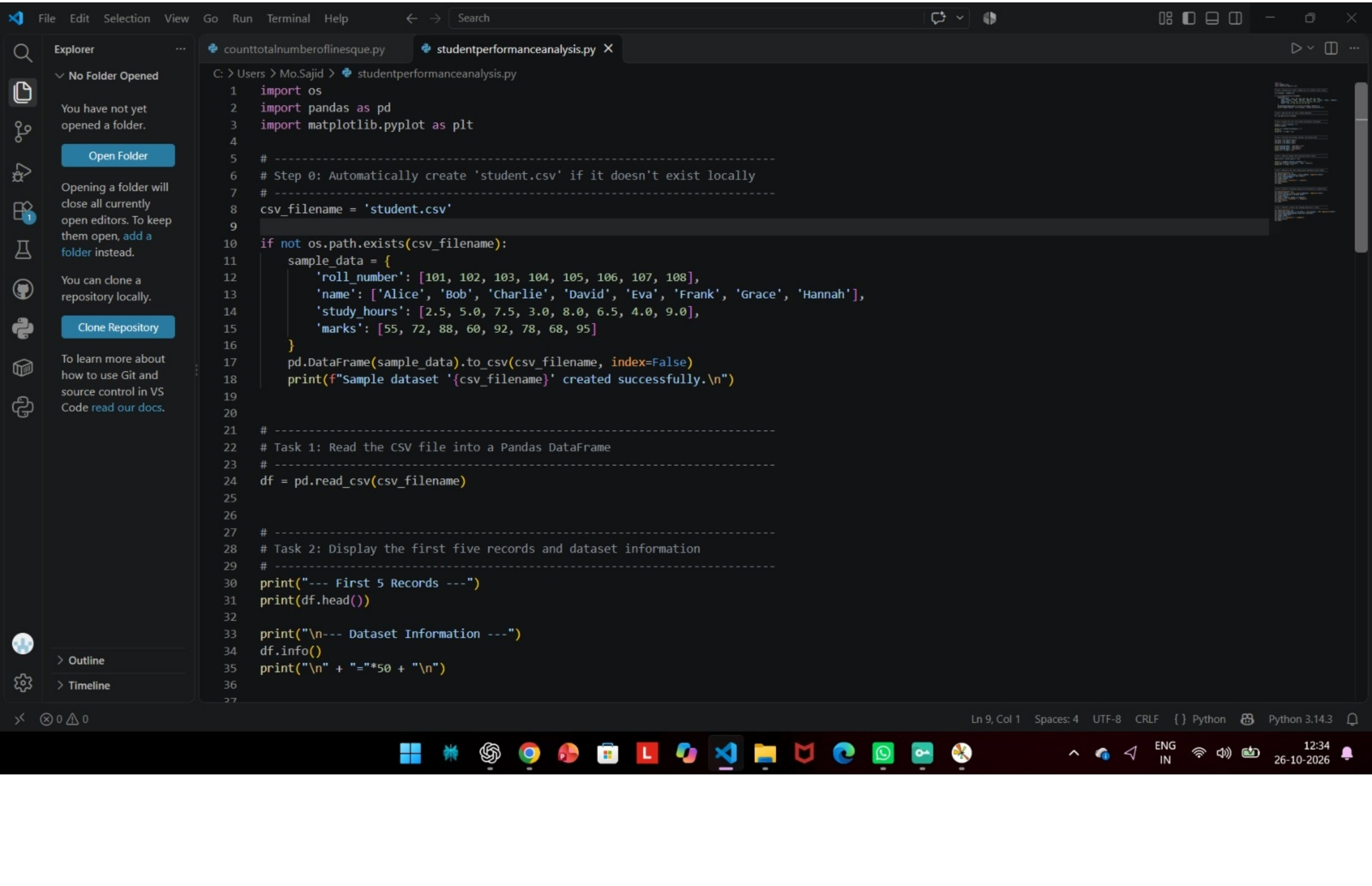
Code

```
PS C:\Users\Mo.Sajid> python -u "c:\Users\Mo.Sajid\counttotalnumberoflinesque.py"
Total number of lines: 5
First two lines written to output.txt
PS C:\Users\Mo.Sajid>
```

0 0

Ln 17, Col 40 Spaces: 4 UTF-8 CRLF { } Python Python 3.14.3

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File Edit Selection View Go Run Terminal Help
Search
Explorer
No Folder Opened
You have not yet opened a folder.
Open Folder
Opening a folder will close all currently open editors. To keep them open, add a folder instead.
You can clone a repository locally.
Clone Repository
To learn more about how to use Git and source control in VS Code read our docs.
Outline
Timeline
counttotalnumberoflinesque.py
studentperformanceanalysis.py
C: > Users > Mo.Sajid > studentperformanceanalysis.py
import os
import pandas as pd
import matplotlib.pyplot as plt
Step 0: Automatically create 'student.csv' if it doesn't exist locally
csv_filename = 'student.csv'
if not os.path.exists(csv_filename):
sample_data = {
'roll_number': [101, 102, 103, 104, 105, 106, 107, 108],
'name': ['Alice', 'Bob', 'Charlie', 'David', 'Eva', 'Frank', 'Grace', 'Hannah'],
'study_hours': [2.5, 5.0, 7.5, 3.0, 8.0, 6.5, 4.0, 9.0],
'marks': [55, 72, 88, 60, 92, 78, 68, 95]
}
pd.DataFrame(sample_data).to_csv(csv_filename, index=False)
print(f"Sample dataset '{csv_filename}' created successfully.\n")
Task 1: Read the CSV file into a Pandas DataFrame
df = pd.read_csv(csv_filename)
Task 2: Display the first five records and dataset information
print("--- First 5 Records ---")
print(df.head())
print("\n--- Dataset Information ---")
df.info()
print("\n" + "="*50 + "\n")
0 0
Ln 9, Col 1 Spaces: 4 UTF-8 CRLF { } Python Python 3.14.3
ENG IN
12:34
26-10-2026

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Outline

Timeline

counttotalnumberoflinesque.py | studentperformanceanalysis.py

C: > Users > Mo.Sajid > studentperformanceanalysis.py

```python
# ------------------------------------------------------------------------
# Task 3: Calculate the average, maximum, and minimum marks
# ------------------------------------------------------------------------
avg_marks = df['marks'].mean()
max_marks = df['marks'].max()
min_marks = df['marks'].min()

print(f"Average Marks : {avg_marks:.2f}")
print(f"Maximum Marks : {max_marks}")
print(f"Minimum Marks : {min_marks}")
print("\n" + "="*50 + "\n")


# ------------------------------------------------------------------------
# Task 4: Identify students who scored more than 75 marks
# ------------------------------------------------------------------------
high_scorers = df[df['marks'] > 75]

print("--- Students Scoring > 75 Marks ---")
print(high_scorers[['roll_number', 'name', 'marks']])
print("\n" + "="*50 + "\n")


# ------------------------------------------------------------------------
# Task 5: Generate a bar chart showing marks obtained by each student
# ------------------------------------------------------------------------
plt.figure(figsize=(8, 5))
plt.bar(df['name'], df['marks'], color='skyblue', edgecolor='black')
plt.title('Marks Obtained by Each Student')
plt.xlabel('Student Name')
plt.ylabel('Marks')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()


```



0 0

Ln 9, Col 1 Spaces: 4 UTF-8 CRLF { } Python Python 3.14.3

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Outline

Timeline

counttotalnumberoflinesque.py

studentperformanceanalysis.py

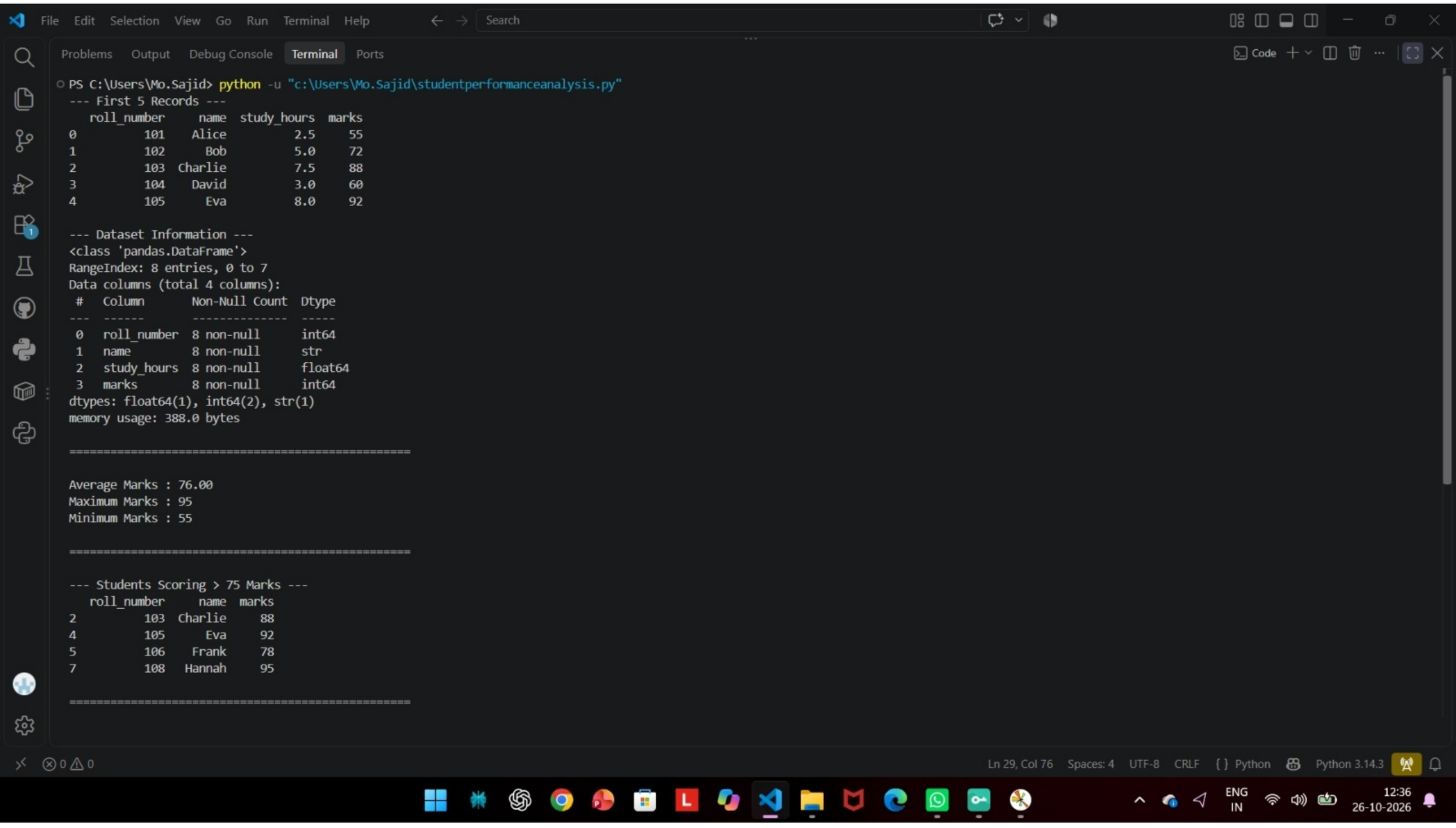
C: > Users > Mo.Sajid > studentperformanceanalysis.py

```
73
74  # ------------------------------------------------------------------------
75  # Task 6: Generate a histogram showing the distribution of student marks
76  # ------------------------------------------------------------------------
77  plt.figure(figsize=(8, 5))
78  plt.hist(df['marks'], bins=5, color='lightgreen', edgecolor='black')
79  plt.title('Distribution of Student Marks')
80  plt.xlabel('Marks')
81  plt.ylabel('Frequency (Number of Students)')
82  plt.grid(axis='y', linestyle='--', alpha=0.7)
83  plt.tight_layout()
84  plt.show()
85
86
87  # ------------------------------------------------------------------------
88  # Task 7: Generate a scatter plot showing study hours vs marks
89  # ------------------------------------------------------------------------
90  plt.figure(figsize=(8, 5))
91  plt.scatter(df['study_hours'], df['marks'], color='purple', s=70, edgecolors='black')
92  plt.title('Relationship Between Study Hours and Marks')
93  plt.xlabel('Study Hours')
94  plt.ylabel('Marks')
95  plt.grid(True, linestyle='--', alpha=0.7)
96  plt.tight_layout()
97  plt.show()
```

0 0

Ln 9, Col 1 Spaces: 4 UTF-8 CRLF { } Python Python 3.14.3

ENG IN 12:34 26-10-2026


File Edit Selection View Go Run Terminal Help
Search
Problems Output Debug Console Terminal Ports
Code
PS C:\Users\Mo.Sajid> python -u "c:\Users\Mo.Sajid\studentperformanceanalysis.py"
--- First 5 Records ---
roll_number name study_hours marks
0 101 Alice 2.5 55
1 102 Bob 5.0 72
2 103 Charlie 7.5 88
3 104 David 3.0 60
4 105 Eva 8.0 92
--- Dataset Information ---
<class 'pandas.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 4 columns):
Column Non-Null Count Dtype
0 roll_number 8 non-null int64
1 name 8 non-null str
2 study_hours 8 non-null float64
3 marks 8 non-null int64
dtypes: float64(1), int64(2), str(1)
memory usage: 388.0 bytes
Average Marks : 76.00
Maximum Marks : 95
Minimum Marks : 55
--- Students Scoring > 75 Marks ---
roll_number name marks
2 103 Charlie 88
4 105 Eva 92
5 106 Frank 78
7 108 Hannah 95
Ln 29, Col 76 Spaces: 4 UTF-8 CRLF { } Python Python 3.14.3
ENG IN
12:36 26-10-2026